

Arman Hossen

Biostatistics · Representation learning · Interpretable models for population and biomedical data

MS Statistics candidate at **Miami University** working at the intersection of **survey-weighted epidemiological inference** and **high-dimensional machine learning**. My work spans two connected strands: nationally representative health surveys (BDHS, MICS, NHANES) analysed with complex-survey methods, and deep representation learning for multi-omics cancer data. What unites them is a single question — **not just what a model predicts, but why, and how much we should trust it.**

[Interactive R Shiny Dashboard](#)

[Website](#)

[Google Scholar](#)

[GitHub](#)

[LinkedIn](#)

[YouTube — Rough Khata](#)

[Curriculum Vitae \(PDF\)](#)

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01

Research at a Glance

Two peer-reviewed publications, an ongoing master's thesis in supervised representation learning, seven manuscripts under review at international journals, and two conference posters.

2

Peer-reviewed articles published
Scientific Reports · Frontiers in Public Health

1

Master's thesis project in deep representation learning
Multi-omics cancer classification, TCGA

7

Manuscripts under review at international journals

2

Conference posters
GSA 2026 · American Heart Association

4

National / consortium data systems analysed
BDHS · MICS · NHANES · TCGA

45k+

Survey and molecular records analysed across projects

TWO ANCHORS OF THE PORTFOLIO

PUBLISHED RESEARCH

Two peer-reviewed articles in *Scientific Reports* and *Frontiers in Public Health*

Survey-weighted analyses of nationally representative Bangladeshi health data — a two-decade decomposition of caesarean section determinants across 26,845 women, and a determinants-plus-machine-learning study of pre-

MASTER'S THESIS PROJECT

Supervised Representation Learning for Multi-Omics Cancer Classification

Converting a published multi-omics autoencoder (JISAE) from an unsupervised reconstruction model into an end-to-end supervised classifier in PyTorch — 99.46% held-out accuracy on TCGA pan-cancer data, benchmarked against

hypertension in 10,981 adults. Both full texts are included with this portfolio.

[Read the summaries ↓](#)

[Sci Rep \(PDF\)](#)

[Front Public Health \(PDF\)](#)

random forest and SVM under an identical protocol. Advised by Dr. Michael J. O'Connell, Miami University.

[See the full project ↓](#)

THREE CONNECTED RESEARCH STRANDS

STRAND I

Survey-weighted population health inference

Complex-survey design, multilevel and multinomial models, zero-inflated count models, cascade and decomposition analysis of nationally representative data from Bangladesh and the United States.

STRAND II

Representation learning for high-dimensional biomedical data

Converting multi-omics autoencoders into end-to-end supervised classifiers; joint versus individual embedding structure; rigorous cross-validated hyperparameter search and honest benchmarking against classical baselines.

STRAND III

Trustworthy prediction: calibration, uncertainty, interpretability

Conformal prediction for screening, fairness and spatial transportability of clinical ML, and variable-importance methods for deep networks — the through-line connecting my applied and methodological work.

RESEARCH PIPELINE — STATUS OF MANUSCRIPTS

2 Published

Scientific Reports · Frontiers PH

7 Under review

Sci Rep · PLOS Digit Health · BMC Preg & Childbirth ×2 · BMC Bioinform

2 In preparation

Thesis + systematic review

2 Conference posters

GSA 2026 · AHA (BCVS)

DATA SYSTEMS USED ACROSS THE PORTFOLIO

BDHS 2004–2021  6 studies · up to 26,845 women

MICS 2019 / 2022  3 studies · child health & care cascades

NHANES  2 studies · cardiometabolic screening & ideal CVH

TCGA pan-cancer  1,866 patients · 21,577 molecular features

Portfolio composition. Bar widths are proportional to the number of studies drawing on each data system; rejected and unsubmitted work is excluded.

02

Peer-Reviewed Publications

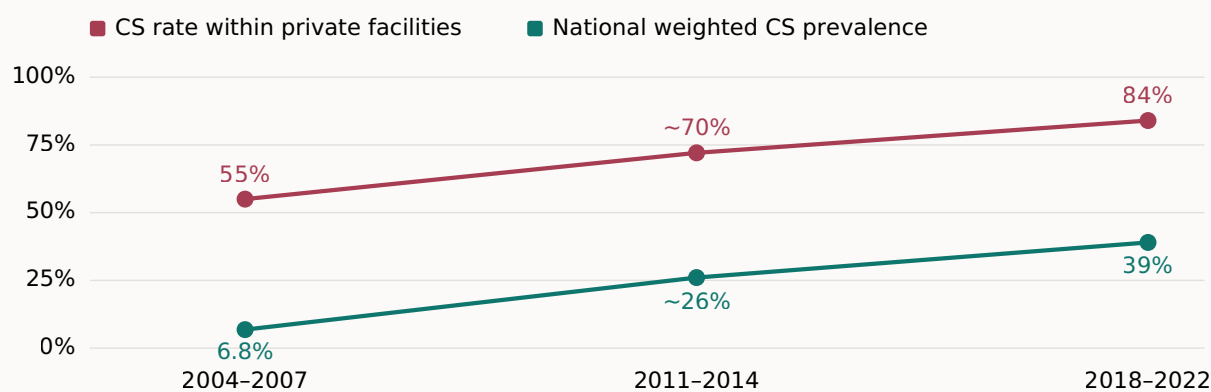
Full-text PDFs are included with this portfolio. My specific methodological contribution is stated for each.

From medical necessity to private-sector dominance: a two-decade evolution of caesarean section determinants in Bangladesh (2004–2022)

Rony, A. R., **Hossen, A.**, Asha, A. S., & Nahin, K. S. A.

A survey-weighted analysis of **26,845 ever-married women** across three BDHS waves (2004–2007, 2011–2014, 2018–2022), harmonising variables and administrative divisions to trace how the determinants of caesarean delivery shifted as national prevalence rose from roughly 4% to 45%.

- Weighted CS prevalence rose from **6.8% (2004–2007) to 39% (2018–2022)**; within private facilities it expanded from ~55% to **84%**.
- Clinical indication weakened as a driver: multiple births fell from AOR 9.55 to **AOR 2.93**, and frequent antenatal care from AOR 9.62 to 4.62 — evidence of saturation beyond medical necessity.
- The socioeconomic gradient inverted: women in the "poorer" quintile moved from AOR 0.53 (protective) to **AOR 1.45**, showing surgical delivery penetrating lower-income groups.
- Coastal residence remained consistently protective relative to the Central region across all three periods.



Survey-weighted BDHS estimates, $n = 26,845$ ever-married women. Mid-period values shown for orientation.

Divergence between national caesarean prevalence and private-sector saturation, 2004–2022.

My contribution: variable harmonisation across three non-identical BDHS waves, survey-weighted multivariable logistic regression accounting for stratification, clustering and sampling weights, comparative interpretation of period-specific adjusted odds ratios, and manuscript drafting and revision.

Full paper (PDF)

DOI: 10.1038/s41598-026-52229-9

Pre-hypertension in Bangladesh: evidence from BDHS 2022

Nahin, K. S. A., & **Hossen, A.**

Determinants of pre-hypertension (JNC-7 definition) among Bangladeshi adults using nationally representative BDHS 2022 data (**n = 10,981** weighted), pairing survey-weighted logistic regression with a random forest model to test whether machine learning adds predictive value over conventional inference. Framed around SDG target 3.4.

- Age and BMI dominated: adults aged ≥ 60 had **AOR 2.75 (95% CI 2.32–3.26)** versus those under 30.
- A **reversed social gradient** — higher education significantly *increased* risk (AOR 1.26), contrary to patterns typical of high-income settings.
- The crude diabetes–pre-hypertension association **disappeared after multivariable adjustment**, a confounding result with direct implications for screening guidance.
- Random forest and logistic regression performed near-identically (**AUC 0.619 vs 0.628**) and produced convergent variable rankings — reported as evidence of stability rather than as a failed ML result, with the interpretable model retained as primary.

DISCRIMINATION: FLEXIBLE MODEL DID NOT BEAT THE INTERPRETABLE ONE

Logistic regression  AUC 0.628

Random forest  AUC 0.619

0.50 (chance)

CONVERGENT VARIABLE IMPORTANCE — TOP PREDICTORS RANKED BY RANDOM FOREST

1 · Age

2 · Body mass index

3 · Sex

— ranking mirrored the independent logistic regression findings

Two methodologically distinct approaches converged on the same risk factors; the added complexity of the ensemble brought no compensating predictive benefit.

My contribution: software implementation, formal analysis, data curation, and visualization for the machine learning component — I built the random forest classifier as a non-parametric robustness check on the primary survey-weighted logistic model, produced the variable-importance ranking, and generated the confusion-matrix and ROC diagnostics.

Full paper (PDF)

DOI: 10.3389/fpubh.2026.1777932

03

Manuscripts Under Review

Seven co-authored manuscripts currently under peer review or in editorial technical check at international journals. Most apply complex-survey methodology to nationally representative data; three extend into machine learning, uncertainty quantification and fairness.

UNDER REVIEW

BMC Bioinformatics

UNDER REVIEW

PLOS Digital Health

Uncertainty-aware multi-outcome screening for undiagnosed cardiometabolic disease in the United States population using NHANES and conformal prediction

Pairs multi-outcome prediction models for diabetes, hypertension, dyslipidemia and chronic kidney disease with **conformal prediction**, so that every screening output carries a calibrated statement of when it should and should not be trusted — addressing the poor calibration typical of machine learning probability outputs in population screening.

A machine learning framework for elective caesarean section in Bangladesh: cluster-aware validation, spatial transportability and fairness across two nationally representative surveys

Cluster-aware cross-validation to avoid optimistic leakage from survey design, out-of-region transportability testing, and explicit fairness auditing across wealth and geography — MICS data.

Rony, A. R., Nahin, K. S. A., Islam, T., Asha, A. S., **Hossen, A.**

UNDER REVIEW

Scientific Reports

Divergent trends in child undernutrition in Bangladesh, 2012–2025: falling stunting alongside rising wasting in three nationally representative surveys

Multi-survey trend decomposition showing that headline progress on stunting conceals a deteriorating acute-malnutrition picture — a result that only appears when indicators are modelled jointly rather than reported separately.

Rony, A. R., Nahin, K. S. A., Islam, T., Asha, A. S., **Hossen, A.**, Sarkar Kripa, A.

UNDER REVIEW

BMC Pregnancy and Childbirth

Where women are lost along the maternal and newborn continuum of care in Bangladesh: a cascade analysis of two nationally representative surveys, 2019 and 2025

Cascade / drop-off modelling locating the specific transitions at which women exit the continuum of care, compared across two MICS rounds.

Nahin, K. S. A., Asha, A. S., **Hossen, A.**, Rony, A. R.

UNDER REVIEW

BMC Pregnancy and Childbirth

Prevalence and determinants of delayed antenatal care initiation among pregnant women in Bangladesh: a nationally representative cross-sectional study

Survey-weighted modelling of timing — rather than mere presence — of first antenatal contact, using BDHS data.

Tasnim, S. J., Rony, A. R., **Hossen, A.**, Asha, A. S., Nahin, K. S. A.

UNDER REVIEW

Humanities and Social Sciences Communications

Socioeconomic inequality and postpartum depression in Bangladesh: evidence from BDHS 2022

Inequality decomposition of postpartum depressive symptoms across the wealth distribution, with survey-weighted regression on the BDHS 2022 mental health module.

Islam, T., Rony, A. R., Nahin, K. S. A., Asha, A. S., **Hossen, A.**, Hossain, M. G.

UNDER REVIEW

Journal of Epidemiology and Global Health

The hypertension care cascade in Bangladesh: a nationally representative analysis of BDHS 2022

Screening → diagnosis → treatment → control cascade quantifying where the largest losses occur in national hypertension management.

Nahin, K. S. A., Rony, A. R., Tasnim, S. J., Islam, T., **Hossen, A.**, Asha, A. S.

POSTER · 2026

Gerontological Society of America Annual Scientific Meeting, National Harbor, MD

Neighborhood influences on mental health in older age: evidence from the 2020 Health and Retirement Study (HRS)

Multilevel modelling of neighbourhood-level context as a determinant of late-life mental health outcomes in a nationally representative US ageing cohort. Abstract ID 2376958.

Ahmed, S. M. F., Asha, A. S., Nahin, K. S. A., & [Hossen, A.](#)

POSTER

American Heart Association — Basic Cardiovascular Sciences Scientific Sessions

Deconstructing sex disparities in ideal cardiovascular health among adults in the United States

Decomposition of the sex gap in ideal cardiovascular health metrics among US adults using NHANES, separating compositional differences from differences in the effect of individual risk components.

Nahin, K. S. A., Asha, A. S., Rony, A. R., & [Hossen, A.](#) · [Abstract](#)

Master's Thesis — Machine Learning Research

My primary methodological research project: converting a published unsupervised multi-omics autoencoder into an end-to-end supervised classifier, and asking what the architecture's structure actually contributes.

ONGOING · MIAMI UNIVERSITY, DEPARTMENT OF STATISTICS

Supervised Representation Learning for Multi-Omics Cancer Classification

The JISAE architecture (Wang & O'Connell, *BMC Bioinformatics*, 2025) compresses paired gene-expression and microRNA data into three embeddings — one shared, two modality-specific — but is trained only to reconstruct, never told the cancer type. My thesis asks: **what happens if the encoder learns directly from the labels?** I reproduced JISAE in full in PyTorch, removed the decoder, attached a classification head trained end-to-end via cross-entropy, and re-optimised every hyperparameter for the new objective.

Advisor: **Dr. Michael J. O'Connell**, Department of Statistics, Miami University · Data: TCGA pan-cancer, 1,866 patients, 6 cancer types, 21,577 molecular features

99.46%	99.58%	0.54%	6.7M	50
Test accuracy, supervised JISAE	Test macro-F1	Train–test gap (no overfitting)	Parameters, 1,494 training patients	Optuna trials × 5-fold stratified CV

METHODOLOGICAL PIPELINE

01

Independent reproduction

Rebuilt the three-branch encoder, preprocessing, and 80/20 stratified split in PyTorch; verified reconstruction loss and downstream accuracy matched the published results.

02

Structured literature review

Identified two established conversion strategies — phased training (OmiVAE, SubOmiEmbed) and joint-loss training (MOSAE) — and chose the phased approach to isolate a single change.

03

Architectural conversion

Replaced the ~21,577-dim reconstruction output with a 6-class head; MSE → cross-entropy; target switched to cancer-type label; search objective flipped to accuracy.

04

Gradient verification

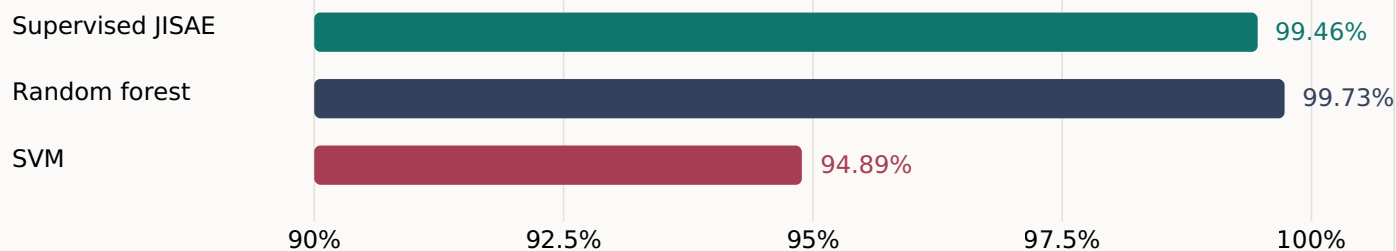
Confirmed classification gradients reach the encoder's earliest layer — the detail distinguishing a genuinely end-to-end model from post-hoc supervised evaluation.

05

Held-out evaluation

Optuna Bayesian search under 5-fold stratified CV inside the training partition only; the 372-patient test set was touched exactly once.

HELD-OUT TEST ACCURACY (372 PATIENTS) — SCALE TRUNCATED AT 90% TO SHOW SEPARATION



GENERALIZATION GAP (TRAIN – TEST) — LOWER IS BETTER



Random forest and SVM baselines were trained on raw concatenated features under the identical train/test split. The converted model misclassified 2 of 372 held-out patients versus 19 for SVM, and generalized far better than SVM despite 6.7 million parameters trained on 1,494 samples.

FULL RESULTS

MODEL	TRAIN ACCURACY	TEST ACCURACY	TRAIN-TEST GAP	TEST MACRO-F1	MISCLASSIFIED
Supervised JISAE (this work)	100.0%	99.46%	0.54%	99.58%	2 / 372
Random forest (raw concatenated features)	100.0%	99.73%	0.27%	99.77%	1 / 372
SVM (raw concatenated features)	99.4%	94.89%	4.51%	94.77%	19 / 372

HONEST INTERPRETATION — WHAT THIS DOES AND DOES NOT SHOW

Every method tested exceeded 94% accuracy, and random forest on entirely raw features was already near-ceiling. The six-class TCGA task therefore offers limited discriminative headroom: molecularly distinct cancer types separate under almost any reasonable classifier. The results establish that **the architectural conversion is technically sound and competitive**, but they do *not* yet establish that JISAE's joint/individual decomposition contributes value beyond a simpler representation. Distinguishing "the conversion works" from "this specific architecture matters" is the open question driving the next phase — and stating that distinction plainly, rather than presenting 99.46% as a clean win, is the part of this work I consider most important.

CURRENT STATUS AND IMMEDIATE NEXT STEPS

Embedding-source ablation

Comparing classification from the joint embedding alone, the modality-specific embeddings alone, and all three combined — a direct test of whether the joint/individual decomposition is functionally contributing. The code already supports it.

Joint-loss variant and a harder dataset

Testing the MOSAE-style combined reconstruction-and-classification loss as a literature-backed alternative to the current phased approach, and moving to a less separable classification task with genuine discriminative headroom.

Interpretability and variable importance

The next phase shifts from predictive performance toward quantifying variable importance inside the supervised network — building toward a visualization of which genes drive predictions at which stage, giving an explanatory dimension to an otherwise black-box architecture.

Related methodological review

A parallel systematic review of machine learning methods for multi-omics cancer classification, mapping the space of architectures and evaluation practices that this thesis sits within.

05

Data Visualization & Reproducible Communication

Analysis that no one can interpret has limited scientific value. Alongside my modelling work I build interactive tools that make statistical results explorable — the same instinct that drives my interest in model interpretability.

FEATURED · LIVE INTERACTIVE APPLICATION

Arman's R Shiny Data Visualization Dashboard

A deployed, publicly accessible R Shiny application built to make statistical distributions, relationships, and survey summaries directly manipulable rather than static. Users adjust inputs and see the visual consequence immediately.

- Built end-to-end in **R / Shiny / ggplot2**, deployed live on shinyapps.io
- Reactive interface — user-driven filtering, variable selection, and plot regeneration
- Designed for teaching and for communicating results to non-statistical audiences
- Extends directly from graduate coursework in **STA 504 Advanced Data Visualization (R)**

[Open the live dashboard →](#)

APPLIED

Real-time monitoring dashboards

As Junior Officer (Data Analytics) at Development Research Initiative, I built **10+ live dashboards** in Power BI and Google Sheets wired to the SurveyCTO API for real-time field-data monitoring, and developed **100+ digital survey instruments** for projects led by the World Bank, Oxford, LSE, Purdue, UPenn and BRAC.

PUBLIC

Statistics teaching channel — *Rough Khata*

I run a public YouTube channel explaining statistical concepts and software workflows, alongside teaching JMP-based labs and holding office hours for **115 undergraduates per semester** as a Graduate Assistant at Miami University. [Visit the channel →](#)

FIGURES

Publication graphics

Visualization was a named contribution on the *Frontiers in Public Health* paper — ROC curves, confusion matrices and variable-importance plots produced in R. Every analysis in this portfolio is scripted and documented for reproduction.

NEXT

Visualization as a research question

My planned thesis extension is explicitly a visualization problem: building an interpretable display of *which genes drive a deep model's predictions at which layer*. Interpretability, for me, is where visualization stops being presentation and becomes method.

Methodological & Technical Profile

Methods listed here have been applied in submitted or published work, not merely studied in coursework.

SURVEY & INFERENCE METHODS

- Complex survey design
- Survey-weighted logistic regression
- Multilevel models
- Multinomial models
- Zero-inflated negative binomial
- Cascade analysis
- Inequality decomposition
- Trend harmonisation across waves

MACHINE LEARNING & DEEP LEARNING

- Autoencoders
- Representation learning
- PyTorch
- Random forest
- SVM
- Optuna / Bayesian HPO
- Stratified k-fold CV
- Cluster-aware validation
- Conformal prediction
- Fairness auditing

STATISTICAL COMPUTING

- R / RStudio
- Python
- SAS
- Stata
- SPSS
- JMP
- Shiny
- ggplot2
- tidyverse
- scikit-learn

DATA ENGINEERING & REPRODUCIBILITY

- Git / GitHub
- Git LFS
- conda environments
- GPU acceleration (CUDA / Apple MPS)
- SurveyCTO
- KoboToolbox
- Power BI
- Scripted audit trails

EDUCATION

DEGREE	INSTITUTION	PERIOD	RESULT
MS, Statistics	Miami University, Oxford, Ohio, USA	Fall 2025 – present	CGPA 3.64 / 4.00
MS, Statistics	Dhaka College (University of Dhaka), Bangladesh	2023 – 2024	CGPA 3.86 – 1st in class
BS (Honours), Statistics	Dhaka College (University of Dhaka), Bangladesh	2018 – 2022	CGPA 3.84 – 1st in class

Graduate coursework at Miami University: STA 502 Statistical Programming (SAS), STA 504 Advanced Data Visualization (R), STA 563 Regression Analysis, STA 566 Experimental Design Methods, STA 664–665 Theory of Statistics I & II.

GRADUATE ASSISTANTSHIP & TEACHING

FALL 2025 – PRESENT

2022 – 2025

Graduate Assistant (Teaching) — Department of Statistics, Miami University

Graduate Assistant for **STA 261 (Statistics)**, supporting course delivery and undergraduate instruction within the department.

- Instruct weekly lab sessions in **JMP**, guiding students through hands-on data analysis exercises.
- Hold office hours and provide academic support to **115 undergraduate students each semester**.
- Fully funded assistantship alongside MS coursework and thesis research.

Tutor — Statistics and Mathematics

Tutored over **200 undergraduate students** across Introduction to Statistics, Probability & Probability Distributions, Linear Algebra, Calculus I & II, and Ordinary Differential Equations.

Teaching at this volume is also where my interest in visualization began — most of what students struggle with is not the arithmetic but the picture behind it.

07

Complete Research Record

All published, under-review and in-preparation work on which I am an author, as of August 2026.

#	TITLE	OUTLET	DATA	STATUS
1	From medical necessity to private-sector dominance: caesarean section determinants in Bangladesh (2004–2022)	Scientific Reports	BDHS	Published
2	Pre-hypertension in Bangladesh: evidence from BDHS 2022	Frontiers in Public Health	BDHS	Published
3	Uncertainty-aware multi-outcome screening for undiagnosed cardiometabolic disease using NHANES and conformal prediction	BMC Bioinformatics	NHANES	Under review
4	A machine learning framework for elective caesarean section: cluster-aware validation, spatial transportability and fairness	PLOS Digital Health	MICS	Under review
5	Divergent trends in child undernutrition in Bangladesh, 2012–2025	Scientific Reports	MICS	Under review
6	Where women are lost along the maternal and newborn continuum of care	BMC Pregnancy and Childbirth	MICS	Under review
7	Prevalence and determinants of delayed antenatal care initiation	BMC Pregnancy and Childbirth	BDHS	Under review
8	Socioeconomic inequality and postpartum depression in Bangladesh	Humanities and Social Sciences Communications	BDHS	Under review
9	The hypertension care cascade in Bangladesh	Journal of Epidemiology and Global Health	BDHS	Under review
10	Neighborhood influences on mental health in older age (HRS 2020)	GSA Annual Scientific Meeting 2026	HRS	Poster
11	Deconstructing sex disparities in ideal cardiovascular health among US adults	American Heart Association (BCVS)	NHANES	Poster
12	Supervised representation learning for multi-omics cancer classification	MS thesis · manuscript planned	TCGA	In progress
13	Systematic review: machine learning methods for multi-omics cancer classification	Target: methods journal	—	In preparation

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[Shiny Dashboard](#) [Website](#) [Google Scholar](#) [GitHub](#) [LinkedIn](#) [YouTube](#) [CV](#)

Research portfolio compiled August 2026. Manuscript statuses current as of this date.